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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/233,324	04/16/2021	Yu Fang	SONA-0006-U01-C08	1095

87084 7590 07/16/2021

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EXAMINER	
LIN, JSING FORNG	

ART UNIT	PAPER NUMBER
2446	

NOTIFICATION DATE	DELIVERY MODE
07/16/2021	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 17/233,324	Applicant(s) Fang et al.	
	Examiner JSING FORNG LIN	Art Unit 2446	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 4/16/2021.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-10 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-10 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 4/16/2021 and 7/1/2021.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

This Action is in response to application filed on 4/16/2021.

Claims 1-10 are pending in this application.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 4/16/2021 and 7/1/2021 are in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the claims at issue are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., In re Berg, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); In re Goodman, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir.

1993); In re Longi, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); In re Van Ornum, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); In re Vogel, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and In re Thorington, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the reference application or patent either is shown to be commonly owned with this application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO internet Web site contains terminal disclaimer forms which may be used. Please visit http://www.uspto.gov/forms/. The filing date of the application will determine what form should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to http://www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

Claims 1-10 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 24-27 of co-pending U.S. Application No. 17/233,319. Although the claims at issue are not identical, they are not patentably distinct from each other because claims in the co-pending application anticipated claims in the present application.

U.S. Application No. 17/233,319	Present Application No. 17/233,324
Claim 24. A system comprising: a vehicle having a first network and a second network; a first device on the first network; and a converged network device (CND) interposed between the first network and the second network and structured to facilitate communications between the first device and the second network; wherein the first network is of a different type than the second network; wherein the first network is a Controller Area Network (CAN) based network and the second network is an Ethernet based network; wherein the CND comprises: a configurable edge gateway (CEG) structured to access the CAN based network; an Ethernet switch structured to access to the Ethernet based network; and wherein the CEG provides a message to the Ethernet switch in response to a corresponding message on the CAN based network.	Claim 1. A system comprising: a vehicle having an Ethernet based network and a controller area network (CAN) based network; a CAN vehicle control device disposed onboard the vehicle, connected to the CAN based network, and structured to control operation of a component of the vehicle; an Ethernet vehicle control device disposed onboard the vehicle, connected to the Ethernet based network, and structured to electrically communicate with the CAN vehicle control device; an Ethernet switch disposed onboard the vehicle and having a plurality of physical ports connected to the Ethernet based network, at least one physical port of the plurality forming part of a physical network path between the Ethernet switch and the Ethernet vehicle control device; a CAN gateway disposed onboard the vehicle and connected to the CAN based

	network and the Ethernet switch; and a network convergence circuit defined at least in part by the Ethernet switch and/or the CAN gateway, and structured to facilitate electronic communications between the Ethernet vehicle control device and the CAN vehicle control device by translating and relaying data messages between the Ethernet based network and the CAN based network

The scope of independent Claim 5 of the instant application is covered by claims 24 and 25 of the co-pending application.

As for the other dependent claims:

The scope of Claims 2-4, 6-10 of the instant application is covered by claims 24-27 of the co-pending application in view of Roese et al (US 2009/0187968 A1) which describe exchange message between external server or mobile device (see paragraphs 0023 and 0047-48).

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole

> would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims 1, 5, 8-10 are rejected under 35 U.S.C. 103 as being unpatentable over SCHOCH (US 2019/0356574 A1 in view of Akella et al (US 2020/0036717 A1).

Regarding claim 1, SCHOCH teaches a system comprising: a vehicle having an Ethernet based network and a controller area network (CAN) based network (SCHOCH fig. 1 [0040] vehicle network with CAN and Ethernet);

a CAN vehicle control device disposed onboard the vehicle, connected to the CAN based network, and structured to control operation of a component of the vehicle (SCHOCH fig. 1 item #21 vehicle control devices to control vehicle operation can be a CAN bus [0040][0043-0044]);

an Ethernet vehicle control device disposed onboard the vehicle, connected to the Ethernet based network, and structured to electrically communicate with the CAN

vehicle control device (SCHOCH [0040][0043-0044] exchange message 23 between different network zone, for example CAN and Ethernet);

an Ethernet switch disposed onboard the vehicle and having a plurality of physical ports connected to the Ethernet based network; a CAN gateway disposed onboard the vehicle and connected to the CAN based network and the Ethernet switch (SCHOCH [0031] domain controller can be Ethernet or CAN gateway / switch; [0028] exchange/translate message between different protocol zones), and

a network convergence circuit defined at least in part by the Ethernet switch and/or the CAN gateway and structured to facilitate electronic communications between the Ethernet vehicle control device and the CAN vehicle control device by translating and relaying data messages between the Ethernet based network and the CAN based network (SCHOCH [0040][0043] exchange message between different network zone, i.e. network zone can be CAN and Ethernet, [0028] control unit translate message between zone).

However, SCHOCH does not explicitly teach an Ethernet switch disposed onboard the vehicle at least one physical port of the plurality forming part of a physical network path between the Ethernet switch and the Ethernet vehicle control device and CAN gateway (Note: SCHOCH disclose domain controller can be Ethernet gateway or router / switch as recited above);

Akella teaches an Ethernet switch disposed onboard the vehicle at least one physical port of the plurality forming part of a physical network path between the Ethernet switch and the Ethernet vehicle control device and CAN gateway (Akella fig. 5 [0050] Ethernet switch interconnect with IP gateway and CAN gateway).

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify SCHOCH in view of Akella to an Ethernet switch disposed onboard the vehicle at least one physical port of the plurality forming part of a physical network path between the Ethernet switch and the Ethernet vehicle control device and CAN gateway.

One of ordinary skill in the art before the effective filing date of the claimed invention would have been motivated because it would have enabled the system to perform network interconnect more efficiently (Akella [0050]).

Regarding claim 5, SCHOCH-Akella teaches an apparatus comprising: a first network interface circuit structured to interpret a first Ethernet data set and to transmit a second Ethernet data set; a second network interface circuit structured to transmit a first controller area network (CAN) data set and to interpret a second CAN data set (SCHOCH [0040][0043-0044]); a translation circuit structured to: determine a first message value from the first Ethernet data set and the first message value in the CAN data set; and determine a second message value from the second CAN data set and the second message value in the second Ethernet data set; wherein the first network interface circuit, the translation circuit, and the second network interface circuit are defined by an Ethernet switch and/or a CAN gateway both structured to be incorporated into a vehicle (SCHOCH [0040][0043] exchange message between different network zone, i.e. can be CAN and Ethernet, [0028] control unit translate message between zone).

However, SCHOCH does not explicit determine a first message value from the first Ethernet data set and to encapsulate the first message value in the CAN data set; and determine a second message value from the second CAN data set and to encapsulate the second message value in the second Ethernet data set;

Akella teaches determine a first message value from the first Ethernet data set and to encapsulate the first message value in the CAN data set; and determine a second message value from the second CAN data set and to encapsulate the second message value in the second Ethernet data set (Akella [0045-0046] encapsulated / de-capsulated CAN to Ethernet and Ethernet to CAN);

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify SCHOCH in view of Akella to determine a first message value from the first Ethernet data set and to encapsulate the first message value in the CAN data set; and determine a second message value from the second CAN data set and to encapsulate the second message value in the second Ethernet data set.

One of ordinary skill in the art before the effective filing date of the claimed invention would have been motivated because it would have enabled the system to perform different type message communication more efficiently (Akella [0044-0045]).

Regarding claim 8, SCHOCH-Akella teaches the apparatus of claim 5, wherein the first network interface circuit is defined by the Ethernet switch and the second network interface circuit is defined by the CAN gateway (SCHOCH [0040], Akella [0050] Ethernet switch interconnect with IP gateway and CAN gateway).

Regarding claim 9, SCHOCH-Akella teaches the apparatus of claim 8, wherein the translation circuit is defined by the CAN gateway (SCHOCH [0028], Akella [0044-0045]).

Regarding claim 10, SCHOCH-Akella teaches the apparatus of claim 8, wherein the translation circuit is defined by the Ethernet switch (SCHOCH [0028]. Akella [0050]).

Claims 2-4, 6-7 are rejected under 35 U.S.C. 103 as being unpatentable over SCHOCH (US 2019/0356574 A1 in view of Akella et al (US 2020/0036717 A1) as applied to claims 1, 5, 8-10 above and further in view of Roese et al (US 2009/0187968 A1).

Regarding claim 2, SCHOCH-Akella teaches the system of claim 1, as recited above.

However, SCHOCH-Akella does not explicit teach wherein the network convergence circuit is selectively configurable via a configuration command value transmitted over the Ethernet based network from a service device external to the vehicle (Note: SCHOCH discloses [0047-0048] exchange message between external server or mobile device; [0012] message includes control command).

Roese teaches wherein the network convergence circuit is selectively configurable via a configuration command value transmitted over the Ethernet based

network from a service device external to the vehicle (Reese [0031] acquire policy from remote server; [0023] assign and modify device based on policy)

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify SCHOCH-Akella in view of Roese wherein the network convergence circuit is selectively configurable via a configuration command value transmitted over the Ethernet based network from a service device external to the vehicle.

One of ordinary skill in the art before the effective filing date of the claimed invention would have been motivated because it would have enabled the system to perform device configuration more efficiently (Roese [0023]).

Regarding claim 3, SCHOCH-Akella-Roese teaches the system of claim 2, wherein: a first physical port of the plurality selectively mirrors a second physical port of the plurality in response to the configuration command value (Roses [0041] port mirroring).

Regarding claim 4, SCHOCH-Akella-Roese teaches the system of claim 2, wherein the configuration circuit is further structured to modify a converged network device (CND) via selectively configuring which of one or more portions of the CND that are defined by the Ethernet switch and/or the CAN gateway (Roese [0023] modify based on received policy, SCHOCH [0028][0032] translate message between first protocol and second protocol based on safety criterion).

Regarding claim 6, SCHOCH-Akella-Roese teaches the apparatus of claim 5 further comprising a configuration circuit defined by the Ethernet switch and/or the CAN gateway and structured to modify the first network interface circuit, the translation circuit, and/or the second network interface circuit in response to a configuration command value (Roese [0023] modify device based on received policy, SCHOCH [0028]).

Regarding claim 7, SCHOCH-Akella-Roese teaches the apparatus of claim 6, wherein the configuration circuit is further structured to selectively configure which of one or more portions of the first network interface circuit, the translation circuit, and/or the second network interface circuit are defined by the Ethernet switch and/or the CAN gateway (Roese [0023] configure devices based on received policy, SCHOCH [0028][0032] domain control may include firewall to specify safety configuration/criterion for protocol translation).

Additional References

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

a. Shalev et al., US 2019/0260800 A1: Cryptic Vehicle Shield.
b. Wu et al., Virtual Controller Area Network.
c. Joshi et al., US 2020/0267080 A1: Systems and methods for data processing and storage in vehicles having a zone based, central computing in-vehicle communication network architecture.

d. Kim et al., US 2016/0255154 A1: Vehicle security network device and design method therefor.

e. Penilla et al., US 2015/0210287 A1: Vehicles and vehicle system for providing access to vehicle controls, functions, environment and applications to guest/passengers vi mobile devices.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEVE LIN whose telephone number is 571-272-5137. The examiner can normally be reached on Monday – Friday 7:30 AM – 5:00 PM.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Brian Gillis can be reached on 571-272-7952. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/S. L./
Examiner, Art Unit 2446

/AUSTIN J MOREAU/
Primary Examiner, Art Unit 2446